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The value of GenAI for peer feedback provision: student perceptions and impacts

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Abstract

Generative Artificial Intelligence (GenAI) has sparked a global debate on its potential as a feedback source for students, yet research in this area remains limited. This study explores students' use of GenAI during peer feedback provision. Fifty-four graduate students enrolled in a master's course in the food science domain at a Dutch university received instruction on the effective and ethical use of GenAI. They then wrote an argumentative essay, provided feedback to peers, and revised their essays. Finally, students completed an online questionnaire regarding their perceptions and use of GenAI for peer feedback provision. Descriptive analyses were applied to survey data and comment data were coded quantitatively for the presence of comment features. The results revealed that just over half of the students chose not to use GenAI for peer feedback provision, primarily because they believed they would learn more by completing the task independently. The remaining students used GenAI to improve both high-level and low-level aspects of their feedback, and most of these students found GenAI to be moderately helpful for peer feedback provision. In terms of its impact on the peer feedback content, students who used GenAI provided more suggestions for high-level issues and offered less mitigating praise for low-level issues compared to those who did not use GenAI for peer feedback provision. These results offer valuable insights for the design and adoption of GenAI tools to enhance peer feedback practices.

Keywords Artificial intelligence, argumentative essay writing, Peer feedback, Generative artificial intelligence, GenAI

Introduction

Writing plays a crucial role in developing academic literacy for university students. Feedback is vital to enhancing students' learning and performance in writing (Hattie & Timperley, 2007). However, instructors face significant challenges in providing effective feedback due to heavy workloads, time constraints, and large class sizes (Shi, 2019). Peer feedback addresses workload concerns. In addition, as a recognized instructional strategy with roots in Vygotsky's sociocultural theory (Vygotsky, 1978), peer feedback is considered especially useful for improving writing and learning (Valero Haro et al., 2024)

by encouraging students to critically review their peers' work, identify problems, and provide suggestions for improvement (Latifi et al., 2021).

As an important learning activity, peer feedback offers various advantages both when students act as feedback receivers and as providers (Gao et al., 2024; Yu, 2019). Feedback from peers can improve students' engagement (Fan & Xu, 2020), motivation (Hsia et al., 2016), and satisfaction with their learning experience (Donia et al., 2022). Peer feedback can also play a crucial role in enhancing students' writing, learning, argumentation, and critical thinking capabilities (e.g., Gielen & De Wever, 2015; Noroozi et al., 2023). For example, in the context of English for Academic Purposes (EAP), research shows positive impacts of peer feedback on improving writing quality and enhancing students' cognitive and emotional states (e.g., Ma, 2020; Wei & Liu, 2024; Zhang et al., 2023). However, the effectiveness of peer feedback is largely dependent on the quality of the feedback that students produce (Kerman et al., 2024). High-quality feedback typically includes a summary of the work, the identification and explanation of the problem, and providing constructive suggestions or solutions for improvement (Wu & Schunn, 2021b).

There are several implementation challenges to productive peer feedback practices in higher education contexts (e.g., Alqassab et al., 2023; Latifi et al., 2021). Some of these challenges stem from psychological, emotional, and social barriers including students' perceptions of their peers and the feedback they receive, concerns about trustworthiness of the feedback, and the tolerance level for receiving and resistance to accepting critical feedback (Latifi et al., 2021; Panadero et al., 2023). Students may also have concerns about losing face or creating potential conflicts with their peers, interpreting critical feedback as personal attacks (Rourke & Kanuka, 2007). Other challenges, such as limited feedback literacy, insufficient content knowledge, and interpersonal dynamics, may hinder the full potential of peer feedback practices in enhancing students' learning (see Banihashem et al., 2024) which could result in surface-level feedback provision (e.g., Gan & Hattie, 2014). Most of these challenges may be linked to students' perceptions of peer feedback (Strijbos et al., 2010; Noroozi et al., 2025). If students perceive their peers lack the knowledge required to identify problems or doubt the effectiveness of input from less competent peers, they may question its quality (Jiang & Yu, 2014). Their perceptions, in turn, may affect their levels of contribution, engagement, motivation, and learning (Chou, 2014; Wu, 2019). A positive attitude and perception enhance effective feedback provision, while a negative attitude may result in the opposite effect and impede students' active engagement (Misiejuk et al., 2021).

To address these challenges, scholars have proposed different frameworks and models (e.g., Panadero & Lipnevich, 2022) and instructional strategies such as training, assessment rubrics, worked examples, etc. (see Farrokhnia et al., 2025; Kerman et al., 2024; Lahza et al., 2025; Latifi et al., 2021) to help enhance student engagement, motivation, and quality in peer feedback practices. Even with all these forms of instructional strategies, not all students can successfully provide high-quality feedback as their domain knowledge and peer feedback literacy could vary among individual learners (see Noroozi et al., 2023).

Rapid development of Generative Artificial Intelligence (GenAI) tools such as ChatGPT has offered ample opportunities to enhance students' learning outcomes by providing feedback that is scalable, timely, and tailored to individual needs (see Deeva et al.,

2021; Debets et al., 2025; Meyer et al., 2024; Su & Yang, 2022). However, using GenAI content without further human input can produce highly erroneous content, entirely missing certain issues or producing plausible content that is in fact false. In addition, even if accurate, if GenAI fully takes over the feedback providing task, then students miss the opportunity to learn from providing feedback.

Therefore, utilizing GenAI within peer feedback setting in the form of so called “hybrid intelligence” could promote collaboration between human (peers) and AI which could potentially contribute to the overall improvement of learning experiences (Banihashem et al., 2025; Li et al., 2024). While peer interactions during peer feedback activities can enhance deeper engagement and reflection, GenAI can serve as a scaffold by guiding students to provide more detailed, effective, and justified feedback (see Banihashem et al., 2024; Er et al., 2024). This human-AI collaboration can lead to more personalized, scalable, and impactful learning experiences regarding peer feedback practices in higher education environments (Hutt et al., 2024).

Recent studies collectively suggest the positive role of GenAI as a valuable tool for supporting and addressing surface-level corrections of the peer feedback message content (e.g., El Alaoui, 2023; Guo, 2024; Hutt et al., 2024; Mahapatra, 2024; Shi & Aryadoust, 2024). It remains unclear how the integration of GenAI into peer feedback provision activities can move beyond simple error correction (grammar, punctuation, etc.) to effectively support higher-order writing skills (Shi & Aryadoust, 2024). Feedback on higher-order writing is particularly important for the learning of both the provider and the receiver (Wu & Schunn, 2020a, b).

Integrating GenAI into peer feedback practices presents challenges, raising questions about whether students can effectively perceive, understand, and utilize GenAI tools as a reliable source of feedback. These challenges include negative perceptions of GenAI, weak GenAI literacy, over-reliance on GenAI tools, and reduced opportunities for creativity. While GenAI offers useful support, this area is still in its early stages, especially for complimenting peer feedback practices (see Banihashem et al., 2024; Er et al., 2024; Farrokhnia et al., 2024), warranting further exploration to better understand how GenAI tools can be effectively integrated into peer feedback processes while addressing these challenges.

Exploring how students perceive and experience the use of GenAI for peer feedback practices is crucial (Banihashem et al., 2024). Equally important is understanding its impacts to determine how GenAI can be effectively integrated to enhance the quality of peer feedback (Pereira et al., 2024). Therefore, this study explores students’ perceptions and experiences with using GenAI to scaffold peer feedback provision while also examining differences in the quality of peer feedback between those who utilize GenAI tools and those who do not.

By addressing this, our study makes three main contributions. (1) It provides empirical insights into how frequently and for what purposes students use GenAI when providing peer feedback. (2) It captures students’ perceptions and experiences with GenAI in this context, including perceived benefits and limitations. (3) It offers a comparative analysis of peer feedback content, focusing on differences in high- and low-level comment features between students who used GenAI and those who did not. Together, these contributions advance the understanding of how GenAI can be used to support peer feedback provision effectively in higher education.

Research questions

This study aims to address the following research questions:

- **Research Question 1:** To what extent do students utilize GenAI for peer feedback provision?
- **Research Question 2:** What are students' perceptions and experiences with using GenAI for peer feedback provision?
- **Research Question 3:** To what extent does the content of the provided peer feedback differ between students who used GenAI and those who did not use it?

Methods

Context and participants

This study was conducted at a Dutch university and involved 54 graduate students enrolled in a master's course in the food science domain. The research setup included three weeks module within the course that was conducted in English, and all students met the university's English language proficiency requirements for admission to an English-taught Master's program (an overall score of 6.5 on the IELTS, 92 on the TOEFL iBT, or a C1 grade on the Cambridge English exam). Of the participants, 50 out of 54 students completed the online post-test questionnaire (female: $N=43$, 86%; male: $N=7$, 15%) and were thus included in this study. Attendance in the module and completion of the assignments were mandatory, as the module was an integral part of the course. However, participation in the research activities, such as completing surveys, was voluntary. All students were fully informed about the research setup, and informed consent was obtained. Students were assured that their personal data would be treated confidentially, with no possibility of linking published results to individual participants. They were also given the freedom to withdraw from the research part of the study and request the deletion of their data at any time. Although students received grades for completing the module as part of the normal practice of the course, their grades were not included in the research analysis. Finally, the study received an ethical approval from the Social Sciences Ethics Committee at the Dutch university, ensuring that all data collection adhered to ethical standards. To support clarity, replicability, and transparency, the structure of this study was informed by the CLOSER checklist (Upsher et al., 2025), which offers a structured framework for reporting quantitative educational intervention research.

Procedure

The course included a three-week module featuring an online peer feedback activity, implemented through FeedbackFruits within the Brightspace learning management system. Before the study began, the course co-instructor (first author of the paper) provided guidelines and materials on writing high-quality argumentative essays, along with approximately 20 min of instruction on using GenAI for argumentative essay writing. This session included an overview of the university's AI policy which emphasizes the legal, ethical, and transparent use of AI, along with practical examples and questions/prompts on how to use GenAI for peer feedback (e.g., identifying strengths and weaknesses in a peer's essay: "I have identified the strengths and weaknesses of my learning partner's essay as follows. Am I missing anything, and why?"). The AI usage policy at the Dutch university encourages the legal, ethical, and transparent use of AI tools to

enhance the quality of teaching and learning; however, the use of AI during exams is prohibited. Students were also instructed not to upload any personal data of their fellow learners into any AI tools.

In the first week of the module, students were then tasked with writing and submitting a draft of an argumentative essay on one of three given controversial topics within the food science domain. In the second week, the students were required to provide written feedback on the essays of two random peers. Neither feedback providers nor receivers were anonymous. Students could access the criteria while commenting on their peers' essays. Despite the annotation function available on the platform, the majority of students provided in-text feedback comments. In the third week, the students revised their essays based on the peer feedback and submitted the revised version of their argumentative essays. Afterward, students completed the online post-test questionnaire about their perceptions and experience using GenAI for the peer feedback provision. In the questionnaire, students were asked to provide their email addresses to link the results with the corresponding students' performance. However, these data were removed permanently later after coding the data. Although students received grades for completing the module as part of the normal practice of the course, their grades were not included in the research analysis.

Measures

Feedback coding process

We analyzed the 100 sets of peer feedback produced by the 50 students who filled out the online post-test questionnaire in two stages. The initial stage involved segmenting the feedback messages into individual comments and coding them according to specific feedback features. We used an event sampling method (Shiffman & Stone, 1998) to segment each set of peer feedback into individual comments for further coding, as most feedback sets addressed multiple issues. An individual feedback comment was defined as one that raises or solves a single issue, resulting in a total of 832 individual comments. To assess segmentation reliability, two of the authors acted as coders and independently coded 85 feedback comments with high reliability ($\kappa = 0.87$). The coders hold relevant educational backgrounds and specific expertise in coding feedback comments. While they were not subject-matter experts in the course content, they possessed the necessary knowledge to accurately identify and classify feedback comments. The first author, who also served as the course co-instructor, provided additional support when coders encountered difficulties related to specific aspects.

Content coding of the feedback messages was conducted using ATLAS.ti. The primary measures of the comments were based on validated coding schemes. The scheme focused on the work of the providers and involved a wide range of comment features, including those related to non-implementable comments (i.e., no revision-oriented information) and implementable features (see Table 1 for definition; see Appendix A for examples). The coding scheme builds on a reliable and detailed framework used in several prior studies (Wu & Schunn, 2020b), but adds metacognitive features (i.e., comments encouraging process reflection). Due to the small data size, metacognitive features, along with problem hedges and solution hedges, have been removed from further analysis. Hedges were defined as specific words or expressions that served to soften the force of statements, aiming to produce more contextually appropriate and polite

Table 1 Feedback features by category, with coding reliability and frequency in the dataset, alongside definitions for presence

Features Category	Feature	Definition
Non Implementable		
Summary ($\kappa = 0.93$, $N = 208$)		Statement of what the writer has done
Pure praise ($\kappa = 0.97$, $N = 259$)		Praise on well-written parts of the essay
Implementable		
Cognitive	Identification ($\kappa = 0.80$, $N = 330$)	The comment identifies the issue.
	Explanation ($\kappa = 0.76$, $N = 342$)	The comment explains the issue.
	Suggestion ($\kappa = 0.84$, $N = 152$)	The comment provides a general suggestion for the issue.
	Solution ($\kappa = 0.84$, $N = 219$)	The comment provides a specific solution to the issue.
Metacognitive ($\kappa = 1.00$, $N = 14$)		Encourages students to evaluate and reflect on their knowledge, skills, or strategies.
Affective	Problem Hedges ($\kappa = 0.86$, $N = 5$)	Hedges used in identification or explanation
	Suggestion Hedges ($\kappa = 0.72$, $N = 42$)	Hedges used in the suggestion
	Mitigating praise ($\kappa = 0.85$, $N = 177$)	Positive feedback is added directly next to negative feedback to soften the criticism.

Table 2 Feedback scope categories, definitions, and examples, with frequency in the data sets

Scope	Definition	Example
High-level ($N = 381$)	Comments about the structure, arguments, and justification of the argumentative essay	Then, after the arguments in favor of your opinion, you need to give a small paragraph about the opposite opinion, a response, and also some evidence for that (acknowledgment, response, and warrant).
Low-level ($N = 451$)	Comments about language and flow	There are some spelling mistakes that can be corrected with a second reading after finishing writing.

language (Fraser, 2010). Problem hedges were expressions used to soften the severity of identified or explained issues, while solution hedges were expressions used to reduce the assertiveness of provided suggestions or solutions.

In the subsequent stage, a second coding scheme was applied, focusing on the scope of each comment in terms of the high- and low-level issues it addressed (Wu & Schunn, 2020a) (see Table 2 for definition and examples, $\kappa = 0.95$). For each student, we categorized their feedback into two groups based on the scope of the issues addressed (i.e., high-level versus low-level issues). We then calculated the average number of each feedback feature in both high-level and low-level comments for each student.

Post-test questionnaire

We developed an online post-test questionnaire to measure students' perceptions and experiences using GenAI tools for the peer feedback provision. The first question asked, "Did you use GenAI tools to support your peer feedback provision during this module?". If the student chose "no", it would be assumed that they did not utilize GenAI tools for peer feedback provision. They would proceed to answer a question that provided

options for reasons for not using GenAI tools and also included an open-ended section for additional responses. If the student chose “yes”, they would be directed to a follow-up question that allowed them to select the GenAI tools they used and also included an open-ended section for additional answers. We then inquired about their experiences using GenAI for peer feedback provision, utilizing items corresponding to various functions of the GenAI on a 4-point Likert scale: Never tried, Not helpful, Moderately helpful, and Very helpful. This data produced two measures: whether GenAI was tried for specific functions (1 = tried, 0 = never tried), and a helpfulness scale: 0 = Not helpful, 1 = Moderately helpful, 2 = Very helpful. Overall helpfulness across items as a single scale produced a highly reliable measure ($\alpha = 0.86$). Two follow-up open-ended questions asked students which aspects of GenAI they found most and least useful, along with their reasons.

Data analysis

Both the analysis of comment data and survey data involved quantitative data analysis. The comment data were turned into quantitative values (presence of features) using the coding procedures described in the analysis section. The survey data were all based upon multiple-choice or check all that apply type questions rather than open text responses. Thus, no thematic analysis was required.

For Research Question 1, we conducted a descriptive analysis to assess the extent to which students utilized GenAI tools for different purposes in their peer feedback provision process. We conducted a paired samples *t*-test on whether students used GenAI at any point during peer feedback. For Research Question 2, a descriptive analysis was performed to provide an overview of students' perceptions and experiences using GenAI tools in peer feedback provision process, including why students did not use GenAI and the perceived helpfulness when they used GenAI.

For Research Question 3, we first divided the students into two groups concerning the use of GenAI in peer feedback provision: Tried GenAI (those who tried at least one GenAI function in their peer feedback provision process, $N = 22$, 44%); Never Tried GenAI (those who never tried any GenAI function in their peer feedback provision process, $N = 28$, 56%). We used the Mann-Whitney U non-parametric test to determine whether the two groups differed in terms of the average number of specific feedback features provided for addressing high- and low-level issues. Four items of our post-test questionnaire directly measured whether students used specific GenAI functions to help them provide specific feedback features (i.e., explanation, identification, balancing the tone of feedback, and constructive suggestions). We again divided students based on whether they used specific GenAI functions for these purposes and conducted a Mann-Whitney U non-parametric test to determine whether the two groups differed in terms of the average number of specific feedback features provided for addressing high- and low-level issues.

Results

Research question 1

In total, 44% of students utilized GenAI for peer feedback provision. In terms of the extent to which students tried different GenAI functions for feedback provision, the most commonly used purposes included checking the grammar and spelling and

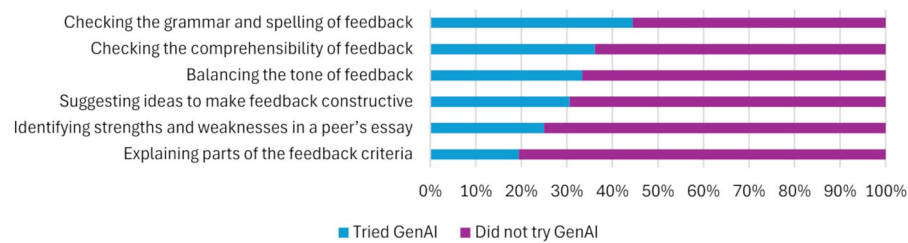


Fig. 1 Percentage of students who tried GenAI functions for peer feedback provision

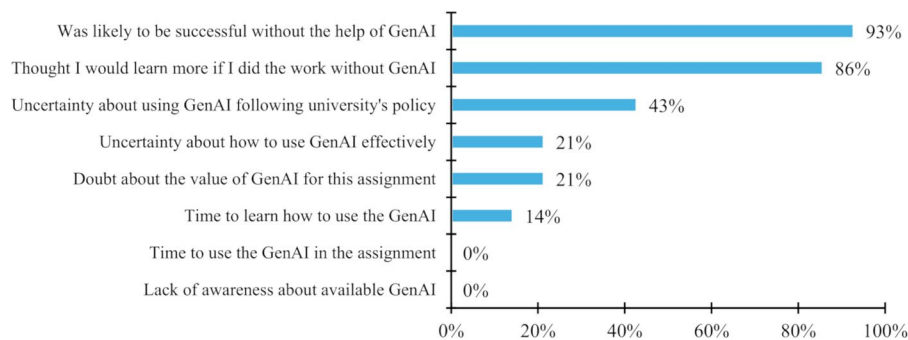


Fig. 2 Reasons for not using GenAI for peer feedback provision

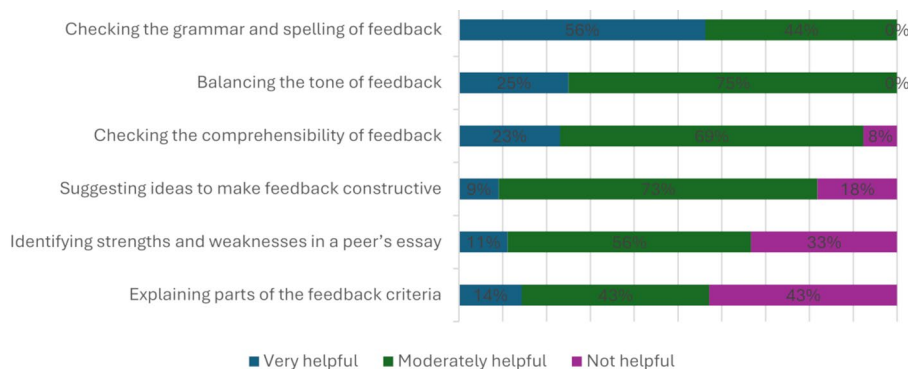


Fig. 3 Student experience with GenAI tools during the peer feedback provision process

checking the comprehensibility of feedback (see Fig. 1), and the lowest percentage of students used GenAI tools to explain parts of the feedback criteria.

Regarding the reasons for not using GenAI during peer feedback provision, most students felt they were likely to succeed without them, and believed they would learn more by completing the work independently (see Fig. 2). Nearly half of the students were uncertain about how to use GenAI tools in compliance with the university's policy. However, none of the students reported lacking awareness about available GenAI tools or not having sufficient time to use them.

Research question 2

Figure 3 presents students' experiences with GenAI tools in the peer feedback provision process. All participants found the tools moderately or very helpful for checking grammar and spelling and balancing the tone of feedback. The purpose of suggesting ideas for

constructive feedback was seen as moderately helpful, while the effectiveness of identifying strengths and weaknesses in peers' essays and explaining feedback criteria varied among students.

In response to the question, "Which use of GenAI did you find most useful and why?" 83% of students highlighted the grammar check and rewriting functionalities. They primarily utilized GenAI tools to enhance their writing by correcting grammatical errors and refining word choices, indicating a reliance on these tools for improving the technical quality of their work. Conversely, when asked, "Which use of GenAI did you find least useful and why?" 33% of students expressed concerns regarding the inadequacy of information provided by the tools. Specifically, they noted that the GenAI outputs often lacked updated information and failed to deliver accurate citations.

Research question 3

Figure 4 presents the descriptive statistics and Mann-Whitney U values, regarding the average number of peer feedback features addressing high- and low-level issues provided by students in two peer feedback sets (see Fig. 4; see Appendix B for more information). Students who used the GenAI tool during peer feedback provision provided significantly more suggestions for high-level issues ($U = 199.5, p < 0.05$) and significantly less mitigating praise for low-level issues ($U = 233, p < 0.05$). Students who used the GenAI tool during peer feedback provision included more identification and explanation features for both high-level and low-level issues compared to those who did not use the tool. However, these differences were not statistically significant.

When considering whether students employed specific functions (i.e., identifying the issues, explaining the criteria, giving constructive suggestions, and balancing the tone) in the feedback provision process, there was only one significant difference in the corresponding feedback features they provided (see Fig. 5, and see Appendix B for more information). When students addressed high level issues, the use of the providing suggestion function led to more suggestions ($U = 105.5, p < 0.05$).

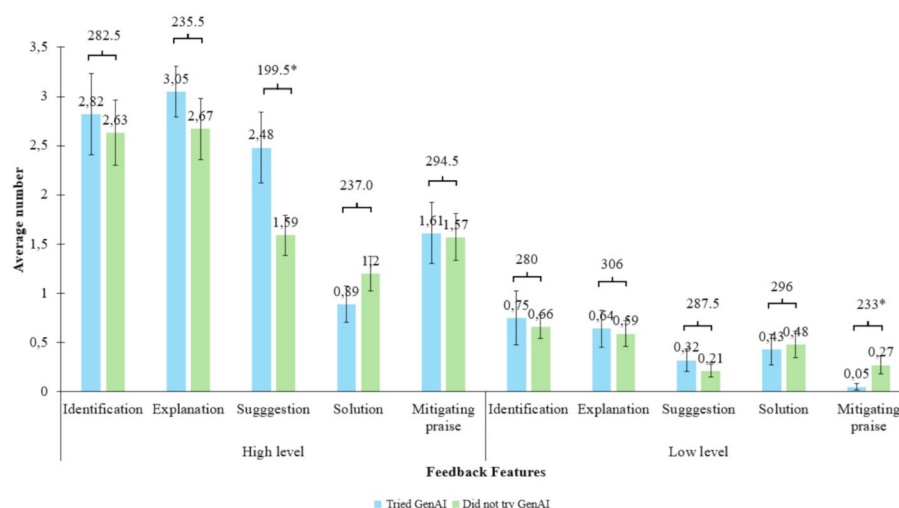


Fig. 4 Average number of peer feedback features within high- and low-level issues, along with U values for the strength of the difference between students who tried and never tried GenAI. *= $p < 0.05$

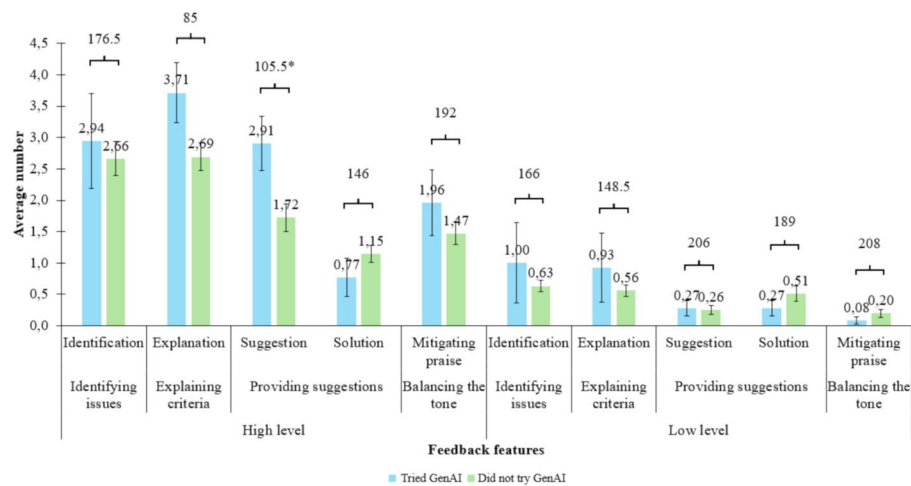


Fig. 5 Mean number of peer feedback features, along with U values for the strength of the difference between students who tried and never tried specific functions with GenAI. * $p < 0.05$

Discussion

This study explored the role of GenAI in scaffolding peer feedback provision within the context of argumentative essay writing. Particularly, we explored how students utilized GenAI during peer feedback provision activities, their perceptions of its use, and how feedback quality differed between students who used GenAI and those who do not use it to provide peer feedback. The findings are discussed below for each research question.

Discussion of the findings of research question 1

The analysis of students' utilization of GenAI during peer feedback provision revealed a mixed pattern, with approximately half of the students actively using GenAI when encouraged to do so. In addition, students especially tended to use GenAI for support on lower-level issues like grammar checking, spell checking, and tone balancing. About half of the GenAI users also used it for higher-level support. The patterns of GenAI use appeared to stem from several factors. First, many students felt they did not need assistance with higher-level issues. Since students are often not aware of the limitations in their own performance quality (Kruger & Dunning, 1999), this is not entirely surprising. Setting higher expectations for peer feedback quality alongside delivering feedback about existing quality and increasing awareness of current peer feedback provision skills might be needed to encourage greater GenAI use.

Second, most students felt the use of GenAI would limit their learning and had some uncertainty about whether using GenAI would violate university policy. These concerns can be cast as potential costs of using GenAI, another key barrier in the adoption of a new technology (Duong et al., 2023). It also highlights the urgent need for improved policy guidelines, as many institutions currently lack formal frameworks for GenAI usage (Ghimire & Edwards, 2024; Michel-Villarreal et al., 2023). Relatively few students were worried about the time invested to use GenAI, another kind of cost. Future research should address ways in which GenAI can be applied without limiting learning from peer feedback or even enhancing learning of the providers.

A third potential factor is perceived usefulness of GenAI, a fundamental determining factor in effective technology use (Davis, 1989). In other contexts, perceived usefulness

(Limna et al., 2023; Shoufan, 2023) and doubts about accuracy (Limna et al., 2023; Schei et al., 2024) often play a fundamental role in determining whether users integrate new technologies into their practices. However, in our study, relatively few students doubted the potential benefits of using GenAI.

Discussion of the findings of research question 2

The study found that alongside the growing integration of GenAI in education (Debets et al., 2025; Samala et al., 2025), a majority of the students in our study found GenAI at least moderately useful for all of the purposes in peer feedback provision. Importantly, most students in our study found it especially helpful for low-level issues (e.g., improving grammar and spelling of feedback). There are several possible explanations for these observations. First, students raised concerns about outdated or inaccurate information and citations provided by GenAI tools, in line with prior research (Wang, 2024). This could be related to their perception of GenAI's usefulness (Davis, 1989). Addressing issues like reliability of the GenAI feedback is fundamental (Kohnke, 2024) to fully realize GenAI tools' contribution in classrooms. Instructors have a significant role in guiding students to maximize GenAI's benefits (Moorhouse & Kohnke, 2024) to support peer feedback provision, although their roles within GenAI-enhanced learning activities have remained underexplored (Lee & Moore, 2024).

The second explanation might be that students' contextual knowledge and critical thinking skills might have made them identify themselves as more effective than GenAI at recognizing deeper contextual issues in essays (Escalante et al., 2023) to provide peer feedback on deeper levels. However, it is fundamental to encourage students to be reflective and not only rely on their contextual knowledge but also integrate useful GenAI features when required (French et al., 2023). Yet, an important caveat is that the use of GenAI may hinder the development of critical thinking, evaluation, and synthesis skills (Bozkurt et al., 2024). The polished, readily available responses can reduce students' problem-solving abilities. GenAI should be used as a scaffold to support students through structured tasks and reflective activities to enhance, rather than replace, meaningful cognitive engagement. While using GenAI, it is important to ensure that students remain critically reflective of its limitations, such as issues of misinformation, hallucinations, and lack of transparency. That is, students need to be equipped with GenAI literacy skills to avoid misinterpreting information or overlooking potential inaccuracies and biases (Bozkurt et al., 2024). Such critical reflection and literacies will better prepare them for the needs of a future society where technology and human insight must coexist synergistically.

The results further suggested a prevalent trend to use popular GenAI tools such as ChatGPT and Grammarly for technical language improvements, which corresponds with GenAI's documented affordances in performing grammatical corrections and paraphrasing (Guo & Wang, 2024) as well as enhancing overall writing accuracy (Mahapatra, 2024; Mizumoto & Eguchi, 2023). These findings highlight the importance of understanding student perceptions to inform policy development and effective integration of GenAI (Liu et al., 2024) also for peer feedback purposes.

Discussion of the findings of research question 3

We found that students utilizing GenAI for peer feedback tended to offer significantly more suggestions for high-level issues (e.g., structure, arguments) but significantly less mitigating praise for low-level issues (e.g., grammar, flow) compared to non-GenAI users. The use of GenAI to provide specific suggestions led to an increase in suggestions for high-level issues, while other features (e.g., identifying issues, explaining criteria, balancing tone) did not show significant effects.

The increased provision of high-level suggestions by GenAI users aligns with the literature indicating that GenAI can scaffold complex cognitive tasks by offering structured prompts or examples for improvement (Hutt et al., 2024; Guo & Wang, 2024). It also highlights GenAI's capacity to support students in tackling higher-order issues beyond simple error correction.

The reduced use of mitigating praise might stem from students assuming that GenAI inherently produces neutral or direct feedback and therefore there is less of a need to soften criticism or maintain interpersonal harmony. Prior studies (Noroozi et al., 2023; Wu & Schunn, 2021a) have noted that mitigating praise serves an important social function in feedback, fostering positive interactions and receptiveness.

A plausible reason for the limited impact of GenAI on other feedback features such as identifying issues or balancing tone might be related to students' GenAI and feedback literacy, indicating gaps in students' understanding of how to fully leverage GenAI for feedback purposes. Students may lack adequate training or guidance to use GenAI comprehensively, resulting in a narrow focus on its suggestion-generation capability (Banihashem et al., 2024).

Another contributing factor could be students' preference for a more human-centered, interpersonal approach to peer feedback. For instance, students who do not use GenAI may prioritize maintaining positive peer relationships by balancing their critiques with mitigating praise, fostering a more supportive and collaborative feedback environment. This behavior aligns with social norms in peer feedback that prioritize politeness and encouragement (Harks et al., 2014; Wu & Schunn, 2021b). In contrast, GenAI users might perceive the tool as reducing their responsibility for the relational aspects of peer feedback, focusing solely on task-related suggestions.

Limitations and implications for future research

As an exploratory study, the relationship between GenAI use and peer feedback features was fundamentally correlational. A future experimental study is needed to test the causal influence of GenAI on peer feedback provision processes and outcomes. The study was also conducted within a single academic context, limiting the generalizability of the findings. Future research could expand the participant pool to include diverse demographics, disciplines, and educational levels to better understand the varied impacts of GenAI for peer feedback practices. This study was conducted within a single peer feedback activity, which might not capture the long-term impact of using GenAI on peer feedback provision. A longitudinal approach would allow researchers to investigate how prolonged exposure to GenAI tools affects student behaviors, perceptions, and learning outcomes during peer feedback activities. This study did not touch upon feedback literacy to explore how students' feedback literacy levels influenced their ability to utilize GenAI effectively for peer feedback provision. Future studies should explore the

interplay between peer feedback literacy and the adoption of GenAI to identify specific students' support needs. This study also only reported on how GenAI can be utilized for peer feedback provision. Future studies could thus focus on peer feedback uptake next to the peer feedback provision. In this study, while we did not directly ask students whether they questioned GenAI feedback more or less than human feedback, our findings suggest that they felt comfortable completing the assignment without significant reliance on GenAI. This may indicate a level of confidence in their ability to provide and evaluate peer feedback independently.

Additionally, this study only focused on exploring students' perceptions and the use of GenAI as scaffolding for peer feedback provision. Future studies should focus on the methodologies of how GenAI and peers should collaborate in order to reach an optimal GenAI-Peer feedback collaboration in the form of so-called "Hybrid Intelligent Feedback" (Banihashem et al., 2025). In addition, research should delve deeper into how the use of GenAI influences socio-emotional aspects of peer feedback, such as trust, relational dynamics, and students' willingness to provide critical and constructive feedback (Huang & Lajoie, 2023). A promising direction for future research would be to investigate the role of trust in AI-generated feedback compared to peer-generated feedback. Specifically, future studies could explore whether students' degree of trust in AI influences the certainty and tone of their feedback, particularly regarding lower-level aspects such as language mechanics. Conducting interviews or focus groups could help to better understand how trust in the feedback source (human vs. AI) affects students' use of hedges and their critical engagement with feedback. Last but not least, given the ethical considerations and uncertainty surrounding GenAI usage for feedback (see Farrokhnia et al., 2024), further studies should focus on developing comprehensive policy frameworks and ethical guidelines to govern the use of such GenAI tools also for peer feedback practices.

Conclusion

This study highlights the potential of GenAI to complement peer feedback provision practices, particularly by enhancing peer feedback quality for supporting surface-level corrections and enhancing high-level peer feedback provision. The overarching results suggest that although GenAI has the potential to support students in providing peer feedback at high levels, integrating it as a complementary tool alongside peer feedback might not be sufficient. Since students mostly seem to use GenAI during peer feedback provision for surface-level corrections, offering structured training and instructional support is recommended to maximize the benefits of GenAI for scaffolding peer feedback provision. Additionally, this study revealed the need for clear institutional guidelines on the ethical and practical use of GenAI to address students' uncertainties about using the tool for learning purposes and to foster meaningful engagement with GenAI-enhanced peer feedback activities. This study also raises the importance of using GenAI for supporting and balancing cognitive and socio-emotional elements of peer feedback, such as mitigating praise. This study also adds value by advocating for a synergistic hybrid GenAI approach, combining human insights with GenAI capabilities, to better prepare students for GenAI-supported peer feedback activities.

Supplementary Information

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Supplementary Material 1

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Author contributions

O. Noroozi led this research experiment. X. Gao and M. Alqassab contributed to the data analysis and writing. C. Schunn contributed to the designing, writing, and reviewing the manuscript. S. Banihashem contributed to the writing and revising the manuscript.

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Data availability

Data will be made available on reasonable request.

Declarations

Competing interests

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References

- Alqassab, M., Strijbos, J. W., Panadero, E., Ruiz, J. F., Warrens, M., & To, J. (2023). A systematic review of peer assessment design elements. *Educational Psychology Review*, 35(1), 18. <https://doi.org/10.1007/s10648-023-09723-7>
- Banihashem, S. K., Kerman, N. T., Noroozi, O., Moon, J., & Drachsler, H. (2024). Feedback sources in essay writing: peer-generated or AI-generated feedback? *International Journal of Educational Technology in Higher Education*, 21(23). <https://doi.org/10.1186/s41239-024-00455-4>
- Banihashem, S. K., Noroozi, O., Khosravi, H., Schunn, C. D., & Drachsler, H. (2025). Pedagogical framework for hybrid intelligent feedback. *Innovations in Education and Teaching International*, 1–17. <https://doi.org/10.1080/14703297.2025.2499174>
- Bozkurt, A., Xiao, J., Farrow, R., Y Bai, J., Nerantz, C., Moore, S. & Asino, T.I. (Ed.). (2024). The manifesto for teaching and learning in a time of generative AI: A critical collective stance to better navigate the future. *Open Praxis* 16 4 487–513 <https://doi.org/10.55982/openpraxis.16.4.777>
- Chou, T. C. R. (2014). A scale of university students' attitudes toward e-learning on the moodle system. *International Journal of Online Pedagogy and Course Design (IJOPCD)*, 4(3), 49–65. <https://doi.org/10.4018/IJOPCD.2014070104>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 319–340. <https://doi.org/10.2307/249008>
- Debets, T., Banihashem, S. K., Joosten-Ten Brinke, D., Vos, T. E., de Buy Wenniger, G. M., & Camp, G. (2025). Chatbots in education: A systematic review of objectives, underlying technology and theory, evaluation criteria, and impacts. *Computers & Education*, 105323. <https://doi.org/10.1016/j.compedu.2025.105323>
- Deeva, G., Bogdanova, D., Serral, E., Snoeck, M., & De Weerd, J. (2021). A review of automated feedback systems for learners: Classification framework, challenges and opportunities. *Computers & Education*, 162, 104094. <https://doi.org/10.1016/j.compedu.2020.104094>
- Donia, M. B., Mach, M., O'Neill, T. A., & Brutus, S. (2022). Student satisfaction with use of an online peer feedback system. *Assessment & Evaluation in Higher Education*, 47(2), 269–283. <https://doi.org/10.1080/02602938.2021.1912286>
- Duong, C. D., Vu, T. N., & Ngo, T. V. N. (2023). Applying a modified technology acceptance model to explain higher education students' usage of chatgpt: A serial multiple mediation model with knowledge sharing as a moderator. *The International Journal of Management Education*, 21(3), 100883. <https://doi.org/10.1016/j.ijme.2023.100883>
- El Alaoui, M. (2023). Improving peer assessment validity and reliability through a fuzzy coherence measure. *IEEE Transactions on Learning Technologies*, 16(6), 892–899. <https://doi.org/10.1109/TLT.2023.3321660>
- Er, E., Akçapınar, G., Bayazit, A., Noroozi, O., & Banihashem, S. K. (2024). Assessing student perceptions and use of instructor versus AI-generated feedback. *British Journal of Educational Technology*, 1–18. <https://doi.org/10.1111/bjet.13558>
- Escalante, J., Pack, A., & Barrett, A. (2023). AI-generated feedback on writing: Insights into efficacy and ENL student preference. *International Journal of Educational Technology in Higher Education*, 20(1), 57. <https://doi.org/10.1186/s41239-023-00425-2>
- Fan, Y., & Xu, J. (2020). Exploring student engagement with peer feedback on L2 writing. *Journal of Second Language Writing*, 50, 100775. <https://doi.org/10.1016/j.jslw.2020.100775>
- Farrokhnia, M., Banihashem, S. K., Noroozi, O., & Wals, A. (2024). A SWOT analysis of chatgpt: Implications for educational practice and research. *Innovations in Education and Teaching International*, 61(3), 460–474. <https://doi.org/10.1080/14703297.2023.2195846>
- Farrokhnia, M., Noroozi, O., Baggen, Y., Biemans, H., & Weinberger, A. (2025). Improving hybrid brainstorming outcomes with computer-supported scaffolds: Prompts and cognitive group awareness. *Computers & Education*, 227, 105229. <https://doi.org/10.1016/j.compedu.2024.105229>
- Fraser, B. (2010). Pragmatic competence: The case of hedging. In G. Kaltenböck, W. Mihatsch, & S. Schneider (Eds.), *New approaches to hedging* (1st ed., pp. 15–34). Emerald Group Publishing Limited.

- French, F., Levi, D., Maczo, C., Simonaityte, A., Triantafyllidis, S., & Varda, G. (2023). Creative use of openai in education: Case studies from game development. *Multimodal Technologies and Interaction*, 7(8), 81. <https://doi.org/10.3390/mti7080081>
- Gan, M. J. S., & Hattie, J. (2014). Prompting secondary students' use of criteria, feedback specificity and feedback levels during an investigative task. *Instructional Science*, 42(6), 861–878. <https://doi.org/10.1007/s11251-014-9319-4>
- Gao, X., Noroozi, O., Gulikers, J., Biemans, H. J., & Banihashem, S. K. (2024). A systematic review of the key components of online peer feedback practices in higher education. *Educational Research Review*, 42, 100588. <https://doi.org/10.1016/j.edurev.2023.100588>
- Ghimire, A., & Edwards, J. (2024). From guidelines to governance: A study of ai policies in education. In *International Conference on Artificial Intelligence in Education* (pp. 299–307). Cham: Springer Nature.
- Gielen, M., & De Wever, B. (2015). Structuring the peer assessment process: A multilevel approach for the impact on product improvement and peer feedback quality. *Journal of Computer Assisted Learning*, 31(5), 435–449. <https://doi.org/10.1111/jcal.12096>
- Guo, K., & Wang, D. (2024). To resist it or to embrace it? Examining chatgpt's potential to support teacher feedback in EFL writing. *Education and Information Technologies*, 29(7), 8435–8463. <https://doi.org/10.1007/s10639-023-12146-0>
- Harks, B., Rakoczy, K., Hattie, J., Besser, M., & Klieme, E. (2014). The effects of feedback on achievement, interest and self-evaluation: The role of feedback's perceived usefulness. *Educational Psychology*, 34(3), 269–290. <https://doi.org/10.1080/01443410.2013.785384>
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112. <https://doi.org/10.3102/003465430298487>
- Hsia, L. H., Huang, I., & Hwang, G. J. (2016). Effects of different online peer-feedback approaches on students' performance skills, motivation and self-efficacy in a dance course. *Computers & Education*, 96, 55–71. <https://doi.org/10.1016/j.compedu.2016.02.004>
- Huang, X., & Lajoie, S. P. (2023). Social emotional interaction in collaborative learning: Why it matters and how can we measure it? *Social Sciences & Humanities Open*, 7(1), 100447. <https://doi.org/10.1016/j.ssaho.2023.100447>
- Hutt, S., DePiro, A., Wang, J., Rhodes, S., Baker, R. S., Hieb, G., & Mills, C. (2024). Feedback on feedback: Comparing Classic Natural Language Processing and Generative AI to Evaluate Peer Feedback. In *Proceedings of the 14th Learning Analytics and Knowledge Conference* (pp. 55–65). <https://doi.org/10.1145/3636555.3636850>
- Jiang, J., & Yu, Y. (2014). The effectiveness of internet-based peer feedback training on Chinese EFL college students' writing proficiency. *International Journal of Information and Communication Technology Education (IJICTE)*, 10(3), 34–46. <https://doi.org/10.4018/IJICTE.2014070103>
- Kerman, N. T., Noroozi, O., Banihashem, S. K., Karami, M., & Biemans, H. J. (2024). Online peer feedback patterns of success and failure in argumentative essay writing. *Interactive Learning Environments*, 32(2), 614–626. <https://doi.org/10.1080/10494820.2022.2093914>
- Kohnke, L. (2024). Exploring EAP students' perceptions of GenAI and traditional grammar-checking tools for language learning. *Computers and Education: Artificial Intelligence*, 7, 100279. <https://doi.org/10.1016/j.caeai.2024.100279>
- Kruger, J., & Dunning, D. (1999). Unskilled and unaware of it: How difficulties in recognizing one's own incompetence lead to inflated self-assessments. *Journal of Personality and Social Psychology*, 77(6), 1121–1134. <https://doi.org/10.1037/0022-3514.77.6.1121>
- Lahza, H. F., Demartini, G., Noroozi, O., Gašević, D., Sadiq, S., & Khosravi, H. (2025). Enhancing peer feedback provision through user interface scaffolding: A comparative examination of scripting and self-monitoring techniques. *Computers & Education*, 105260. <https://doi.org/10.1016/j.compedu.2025.105260>
- Latifi, S., Noroozi, O., & Talaee, E. (2021). Peer feedback or peer feedforward? Enhancing students' argumentative peer learning processes and outcomes. *British Journal of Educational Technology*, 52(2), 768–784. <https://doi.org/10.1111/bjet.13054>
- Lee, S. S., & Moore, R. L. (2024). Harnessing generative AI (GenAI) for automated feedback in higher education: A systematic review. *Online Learning*, 28(3), 82–106. <https://doi.org/10.24059/olj.v28i3.4593>
- Li, Z., Liang, C., Peng, J., & Yin, M. (2024). The value, benefits, and concerns of Generative AI-Powered assistance in writing. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (pp. 1–25). <https://doi.org/10.1145/3613904.3642625>
- Limna, P., Kraiwatit, T., Jangjarat, K., Klayklung, P., & Chocksathaporn, P. (2023). The use of ChatGPT in the digital era: Perspectives on chatbot implementation. *Journal of Applied Learning and Teaching*, 6(1), 64–74. <https://doi.org/10.37074/jalt.2023.6.1.32>
- Liu, Y., Park, J., & McMin, S. (2024). Using generative artificial intelligence/chatgpt for academic communication: Students' perspectives. *International Journal of Applied Linguistics*, 34(4), 1437–1461. <https://doi.org/10.1111/ijal.12574>
- Ma, Q. (2020). Examining the role of inter-group peer online feedback on Wiki writing in an EAP context. *Computer Assisted Language Learning*, 33(3), 197–216. <https://doi.org/10.1080/09588221.2018.1556703>
- Mahapatra, S. (2024). Impact of ChatGPT on ESL students' academic writing skills: A mixed methods intervention study. *Smart Learning Environments*, 11(9), 1–18. <https://doi.org/10.1186/s40561-024-00295-9>
- Meyer, J., Jansen, T., Schiller, R., Liebenow, L. W., Steinbach, M., Horbach, A., & Fleckenstein, J. (2024). Using LLMs to bring evidence-based feedback into the classroom: AI-generated feedback increases secondary students' text revision, motivation, and positive emotions. *Computers and Education: Artificial Intelligence*, 6, 100199. <https://doi.org/10.1016/j.caeai.2023.100199>
- Michel-Villarreal, R., Vilalta-Perdomo, E., Salinas-Navarro, D. E., Thierry-Aguilera, R., & Gerardou, F. S. (2023). Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences*, 13(9), 856. <https://doi.org/10.3390/educsci13090856>
- Misiejuk, K., Wasson, B., & Egelandsdal, K. (2021). Using learning analytics to understand student perceptions of peer feedback. *Computers in Human Behavior*, 117, 106658. <https://doi.org/10.1016/j.chb.2020.106658>
- Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2), 100050. <https://doi.org/10.1016/j.rmal.2023.100050>
- Moorhouse, B. L., & Kohnke, L. (2024). The effects of generative AI on initial language teacher education: The perceptions of teacher educators. *System*, 122, 103290. <https://doi.org/10.1016/j.system.2024.103290>
- Noroozi, O., Banihashem, S. K., Biemans, H. J., Smits, M., Vervoort, M. T., & Verbaan, C. L. (2023). Design, implementation, and evaluation of an online supported peer feedback module to enhance students' argumentative essay quality. *Education and Information Technologies*, 28(10), 12757–12784. <https://doi.org/10.1007/s10639-023-11683-y>

- Noroozi, O., Alqassab, M., Taghizadeh Kerman, N., Banihashem, S. K., & Panadero, E. (2025). Does perception mean learning? Insights from an online peer feedback setting. *Assessment and Evaluation in Higher Education*, 50(1), 83–97. <https://doi.org/10.1080/02602938.2024.2345669>
- Panadero, E., & Lipnevich, A. A. (2022). A review of feedback models and typologies: Towards an integrative model of feedback elements. *Educational Research Review*, 35, 100416. <https://doi.org/10.1016/j.edurev.2021.100416>
- Panadero, E., Alqassab, M., Fernández Ruiz, J., & Ocampo, J. C. (2023). A systematic review on peer assessment: Intrapersonal and interpersonal factors. *Assessment & Evaluation in Higher Education*, 48(8), 1053–1075. <https://doi.org/10.1080/02602938.2023.2164884>
- Pereira, R., Reis, I. W., Ulbricht, V., & Santos, N. D. (2024). Generative artificial intelligence And academic writing: An analysis of the perceptions of researchers in training. *Management Research: Journal of the Iberoamerican Academy of Management*. <https://doi.org/10.1108/MRIAM-01-2024-1501>
- Rourke, L., & Kanuka, H. (2007). Barriers to online critical discourse. *International Journal of Computer-Supported Collaborative Learning*, 2, 105–126. <https://doi.org/10.1007/s11412-007-9007-3>
- Samala, A. D., Rawas, S., Wang, T., Reed, J. M., Kim, J., Howard, N. J., & Ertz, M. (2025). Unveiling the landscape of generative artificial intelligence in education: A comprehensive taxonomy of applications, challenges, and future prospects. *Education and Information Technologies*, 30(3), 3239–3278. <https://doi.org/10.1007/s10639-024-12936-0>
- Schei, O. M., Møgelvang, A., & Ludvigsen, K. (2024). Perceptions and use of AI chatbots among students in higher education: A scoping review of empirical studies. *Education Sciences*, 14(8), 922. <https://doi.org/10.3390/educsci14080922>
- Shi, M. (2019). The effects of class size and instructional technology on student learning performance. *The International Journal of Management Education*, 17(1), 130–138. <https://doi.org/10.1016/j.ijme.2019.01.004>
- Shi, H., & Aryadoust, V. (2024). A systematic review of AI-based automated written feedback research. *ReCALL*, 1–23. <https://doi.org/10.1017/S0958344023000265>
- Shiffman, S., & Stone, A. A. (1998). Ecological momentary assessment: A new tool for behavioral medicine research. *Technology and Methods in Behavioral Medicine*, 117–131.
- Shoufan, A. (2023). Exploring students' perceptions of chatgpt: Thematic analysis and follow-up survey. *IEEE Access: Practical Innovations, Open Solutions*, 11, 38805–38818. <https://doi.org/10.1109/ACCESS.2023.3268224>
- Srijbos, J. W., Narciss, S., & Dünnebie, K. (2010). Peer feedback content and sender's competence level in academic writing revision tasks: Are they critical for feedback perceptions and efficiency? *Learning and Instruction*, 20(4), 291–303. <https://doi.org/10.1016/j.learninstruc.2009.08.008>
- Su, J., & Yang, W. (2022). Artificial intelligence in early childhood education: A scoping review. *Computers and Education: Artificial Intelligence*, 3, 100049. <https://doi.org/10.1016/j.caeai.2022.100049>
- Upsher, R., Dommett, E., Carlisle, S., Conner, S., Codina, G., Nobili, A., & Byrom, N. C. (2025). Improving reporting standards in quantitative educational intervention research: Introducing the CLOSER and CIDER checklists. *Journal of New Approaches in Educational Research*, 14(1), 1–37. <https://doi.org/10.1007/s44322-024-00022-9>
- Valero Haro, A., Noroozi, O., Biemans, H. J., Mulder, M., & Banihashem, S. K. (2024). How does the type of online peer feedback influence feedback quality, argumentative essay writing quality, and domain-specific learning? *Interactive Learning Environments*, 32(9), 5459–5478. <https://doi.org/10.1080/10494820.2023.2215822>
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.
- Wang, C. (2024). Exploring students' generative AI-assisted writing processes: Perceptions and experiences from native and nonnative english speakers. *Technology Knowledge and Learning*, 1–22. <https://doi.org/10.1007/s10758-024-09744-3>
- Wei, Y., & Liu, D. (2024). Incorporating peer feedback in academic writing: A systematic review of benefits and challenges. *Frontiers in Psychology*, 15, 1506725. <https://doi.org/10.3389/fpsyg.2024.1506725>
- Wu, Z. (2019). Lower english proficiency means poorer feedback performance? A mixed-methods study. *Assessing Writing*, 41, 14–24. <https://doi.org/10.1016/j.asw.2019.05.001>
- Wu, Y., & Schunn, C. D. (2020a). From feedback to revisions: Effects of feedback features and perceptions. *Contemporary Educational Psychology*, 60, 101826. <https://doi.org/10.1016/j.cedpsych.2019.101826>
- Wu, Y., & Schunn, C. D. (2020b). When peers agree, do students listen? The central role of feedback quality and feedback frequency in determining uptake of feedback. *Contemporary Educational Psychology*, 62, 101897. <https://doi.org/10.1016/j.cedpsych.2020.101897>
- Wu, Y., & Schunn, C. D. (2021a). The effects of providing and receiving peer feedback on writing performance and learning of secondary school students. *American Educational Research Journal*, 58(3), 492–526. <https://doi.org/10.3102/0002831220945266>
- Wu, Y., & Schunn, C. D. (2021b). From plans to actions: A process model for why feedback features influence feedback implementation. *Instructional Science*, 49(3), 365–394. <https://doi.org/10.1007/s11251-021-09546-5>
- Yu, S. (2019). Learning from giving peer feedback on postgraduate theses: Voices from master's students in the Macau EFL context. *Assessing Writing*, 40, 42–52. <https://doi.org/10.1016/j.asw.2019.03.004>
- Zhang, F., Schunn, C., Chen, S., Li, W., & Li, R. (2023). EFL student engagement with giving peer feedback in academic writing: A longitudinal study. *Journal of English for Academic Purposes*, 64, 101255. <https://doi.org/10.1016/j.jeap.2023.101255>

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